Ex.no:4B	**EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION**

Aim:

To perform data inspection, conditional filtering, sorting, statistical analysis, grouping, and data visualization on the Daily Google Search Trends US dataset using Pandas and Matplotlib.

Algorithm:

1. Start.
2. Import the required Python libraries: pandas, numpy, matplotlib.pyplot, and seaborn.
3. Load the Daily Google Search Trends US dataset into a pandas DataFrame using pd.read_csv().
4. Inspect the basic structure, initial records, shape, column names, data types, dataset information, descriptive statistics, and missing values.
5. Clean the dataset by handling missing values and duplicate records.
6. Filter the data based on suitable conditions and display the filtered records.
7. Sort the dataset based on search volume in descending order.
8. Compute statistical metrics for search volume, including mean, median, mode, and standard deviation.
9. Group the data by categories and calculate the average search volume using df.groupby().
10. Generate suitable plots, such as bar plots and histograms, to visualize the search-trend analysis.

Program:

```
import os

import glob

import zipfile

import urllib.request

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

DATASET_URL =
"https://www.kaggle.com/api/v1/datasets/download/adrianjuliusaluoch/daily-
google-search-trends-us"

ZIP_PATH = "/content/google_trends_kaggle.zip"
```

```
DATA_DIR = "/content/google_trends_kaggle"
if not os.path.exists(DATA_DIR):
    urllib.request.urlretrieve(DATASET_URL, ZIP_PATH)
    os.makedirs(DATA_DIR, exist_ok=True)
    with zipfile.ZipFile(ZIP_PATH, "r") as z:
        z.extractall(DATA_DIR)
csv_files = glob.glob(os.path.join(DATA_DIR, "**", "*.csv"), recursive=True)
if not csv_files:
    raise FileNotFoundError("Kaggle CSV was not found after download.")
CSV_PATH = csv_files[0]
df = pd.read_csv(CSV_PATH)
print("Kaggle Google Trends dataset loaded successfully.")
print("CSV:", CSV_PATH)
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
display(df.head())
df.columns = df.columns.astype(str).str.strip().str.lower().str.replace(" ",
"_", regex=False)
def find_col(names):
    for name in names:
        if name in df.columns:
            return name
    return None
query_col = find_col(["query", "trends", "trend", "search_query", "keyword",
"term", "search_term"])
date_col = find_col(["date", "collection_date", "start_time",
"date_recorded", "datetime", "timestamp", "time"])
location_col = find_col(["location", "country", "region", "geo"])
volume_col = find_col(["search_volume_lower", "search_volume", "volume"])
if query_col is None:
    raise ValueError("Search-query column not found.")
if date_col is not None:
    df["date"] = pd.to_datetime(df[date_col], errors="coerce")
    df["year"] = df["date"].dt.year
else:
    df["date"] = pd.NaT
    df["year"] = np.nan
df["query"] = df[query_col].astype(str).str.strip()
```

```
def to_volume(value):
    if pd.isna(value):
        return np.nan
    s = str(value).strip().upper().replace(",", "").replace("+", "")
    try:
        if s.endswith("B"): return float(s[:-1]) * 1000000000
        if s.endswith("M"): return float(s[:-1]) * 1000000
        if s.endswith("K"): return float(s[:-1]) * 1000
        return float(s)
    except ValueError:
        return np.nan
if volume_col is not None:
    df["search_volume"] = df[volume_col].apply(to_volume)
else:
    df["search_volume"] = df.groupby("query")["query"].transform("count").astype(float)
df["location"] = df[location_col].astype(str).str.strip() if location_col is not None else "United States"
df["year"] = pd.to_numeric(df["year"], errors="coerce")
df = df.reset_index(drop=True)
print("Prepared columns: query, date, year, location, search_volume")
display(df[["query", "date", "year", "location", "search_volume"]].head())
print("Shape:", df.shape)
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
print("Data Types:")
print(df.dtypes)
recent = df[df["year"] > 2020]
print("Records after 2020:", len(recent))
display(recent[["query", "date", "year", "location", "search_volume"]].head(20))
us = df[df["location"].str.lower().isin(["united states", "us", "usa", "united states of america"])]
print("US records:", len(us))
display(us[["query", "date", "location", "search_volume"]].head(20))
sorted_df = df.sort_values("search_volume", ascending=False, na_position="last")
display(sorted_df[["query", "date", "year", "location", "search_volume"]].head(20))
```

```
v = df["search_volume"].dropna()
print("Mean =", v.mean())
print("Median =", v.median())
m = v.mode()
print("Mode =", m.iloc[0] if len(m) else "No mode")
print("Standard Deviation =", v.std())
display(df["search_volume"].describe().to_frame().T)
display(df.groupby("year")["search_volume"].mean().sort_index().to_frame("Av
erage_Search_Volume"))
display(df.groupby("year")["query"].count().sort_index().to_frame("Number_of
_Trends"))
print("Unique queries:", df["query"].nunique())
print("Year range:", df["year"].min(), "to", df["year"].max())
print("Mean volume:", round(df["search_volume"].mean(), 2))
display(sorted_df[["query", "search_volume"]].head(10))
```

Output:

```
Kaggle Google Trends dataset loaded successfully.
CSV: /content/google_trends_kaggle/trending_searches_in_us.csv
Shape: (31804, 8)
Columns: ['query', 'start_date', 'end_date', 'active', 'search_volume', 'increase_percentage', 'categories', 'trend_b
eakdown']
```

	query	start_date	end_date	active	search_volume	increase_percentage	categories	trend_breakdown
0	shedeur sanders	2026-03-30 18:40:00+00:00	NaN	True	200	50	Sports	NaN
1	whoopi goldberg	2026-03-30 18:40:00+00:00	NaN	True	5000	50	Politics	america first award, mike johnson
2	mo williams	2026-03-30 18:00:00+00:00	NaN	True	5000	300	Sports	mason williams basketball, mason williams
3	sloane stephens	2026-03-30 17:50:00+00:00	NaN	True	1000	200	Sports, Health	NaN
4	canceled tv shows 2026	2026-03-30 17:50:00+00:00	NaN	True	5000	200	Entertainment	renewed and cancelled tv shows 2026

```
Prepared columns: query, date, year, location, search_volume
```

	query	date	year	location	search_volume
0	shedeur sanders	NaT	NaN	United States	200.0
1	whoopi goldberg	NaT	NaN	United States	5000.0
2	mo williams	NaT	NaN	United States	5000.0
3	sloane stephens	NaT	NaN	United States	1000.0
4	canceled tv shows 2026	NaT	NaN	United States	5000.0

```
Shape: (31804, 11)
Rows: 31804
Columns: 11
```

```
query                         object
start_date                    object
end_date                      object
active                          bool
search_volume                float64
increase_percentage            int64
categories                    object
trend_breakdown               object
date                  datetime64[ns]
year                         float64
location                      object
dtype: object

US records: 31804
```

	query	date	location	search_volume
0	shedeur sanders	NaT	United States	200.0
1	whoopi goldberg	NaT	United States	5000.0
2	mo williams	NaT	United States	5000.0
3	sloane stephens	NaT	United States	1000.0
4	canceled tv shows 2026	NaT	United States	5000.0
5	atlas - guadalajara	NaT	United States	50000.0
6	karoline leavitt	NaT	United States	2000.0
7	kali uchis	NaT	United States	2000.0
8	germany vs ghana	NaT	United States	2000.0
9	caleb flynn	NaT	United States	1000.0
10	lindsey graham	NaT	United States	2000.0
11	mohammad bagher ghalibaf	NaT	United States	200.0
12	national burrito day	NaT	United States	10000.0
13	arc raiders	NaT	United States	2000.0
14	cook county treasurer	NaT	United States	500.0
15	carlie irsay-gordon	NaT	United States	1000.0
16	john groce	NaT	United States	500.0
17	finley bizjack	NaT	United States	200.0
18	topps	NaT	United States	500.0
19	yulia putintseva	NaT	United States	200.0

	query	date	year	location	search_volume
5442	chuck norris	NaT	NaN	United States	10000000.0
25024	savannah guthrie	NaT	NaN	United States	10000000.0
26437	catherine o'hara	NaT	NaN	United States	10000000.0
24439	2026 winter olympics	NaT	NaN	United States	10000000.0
22528	bad bunny halftime show	NaT	NaN	United States	10000000.0
19529	2026 winter olympics alpine skiing	NaT	NaN	United States	5000000.0
28538	rams vs seahawks	NaT	NaN	United States	5000000.0
22028	bad bunny	NaT	NaN	United States	5000000.0
23392	what time is the super bowl	NaT	NaN	United States	5000000.0
12797	2026 winter paralympics	NaT	NaN	United States	5000000.0
15314	iran	NaT	NaN	United States	5000000.0
29481	patriots vs broncos	NaT	NaN	United States	5000000.0
18427	2026 winter olympics women single skating free...	NaT	NaN	United States	5000000.0
31715	miami fl vs indiana	NaT	NaN	United States	5000000.0
26949	superbowl sunday 2026	NaT	NaN	United States	5000000.0
5806	men's march madness	NaT	NaN	United States	5000000.0
22903	what time does the super bowl start	NaT	NaN	United States	5000000.0
31020	winter storm watch	NaT	NaN	United States	5000000.0
17464	2026 winter olympics	NaT	NaN	United States	5000000.0
22648	patriots vs seahawks	NaT	NaN	United States	5000000.0

```
Mean = 27489.891208653
Median = 2000.0
Mode = 2000.0
Standard Deviation = 219937.68485987393
```

	Average_Search_Volume
year	

	Number_of_Trends
year	

	count	mean	std	min	25%	50%	75%	max
search_volume	31804.0	27489.891209	219937.68486	100.0	500.0	2000.0	10000.0	10000000.0

```
Unique queries: 19709
Year range: nan to nan
Mean volume: 27489.89
```

	query	search_volume
5442	chuck norris	10000000.0
25024	savannah guthrie	10000000.0
26437	catherine o'hara	10000000.0
24439	2026 winter olympics	10000000.0
22528	bad bunny halftime show	10000000.0
19529	2026 winter olympics alpine skiing	5000000.0
28538	rams vs seahawks	5000000.0
22028	bad bunny	5000000.0
23392	what time is the super bowl	5000000.0
12797	2026 winter paralympics	5000000.0

Result: